

Building Orbiter Sample Programs (Linux)

September 27, 2026

1 Getting everything ready

The samples are built with CMake from a terminal, with the same compiler and libraries as Orbiter itself. On Ubuntu/Debian, install these packages (other distributions have them under their own names):

```
sudo apt install cmake ninja-build g++ qt6-base-dev python3
```

- **cmake** and **ninja-build**: the build system. Ninja is optional, CMake's default generator (make) works too.
- **g++**: a C++20 compiler.
- **qt6-base-dev**: module dialogs and drawing use Qt 6 on Linux.
- **python3**: dialog resources (.rc files) are compiled to C++ by *cmake/rc2cpp.py*.

Everything else comes with Orbiter: the Orbiter folder (the *root* folder, which contains the *Orbiter* executable) has the CMake script *CMakeLists.txt*, the *cmake* folder, and the SDK in *Orbitersdk* (headers in *include*, libraries in *lib* and the sample sources in *samples*).

2 Building the samples

Open a terminal in the Orbiter *root* folder and run:

```
cmake -S . -B build -G Ninja
cmake --build build
```

The first command configures the build in the *build* subfolder (leave out `-G Ninja` to use make). The second one builds all samples. The default configuration is *Release*; for debugging, configure with:

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=Debug
```

As on Windows, the CMake scripts use your Orbiter folder as the output directory. The modules are built directly into *Modules*, *Modules/Plugin* and *Modules/Startup*, replacing the modules that came with Orbiter, so no separate install step is needed. Only the intermediate files go to the *build* folder. Keep a copy of your Orbiter folder if you want to go back to the original modules.

To build a single sample, name its target, for example:

```
cmake --build build --target DeltaGlider
```

Plugin modules (MFDs and tools in *Modules/Plugin*) must be activated in the *Modules* tab of the Orbiter Launchpad before they are loaded.

3 Working on the samples

The sample sources are in *Orbitersdk/samples*. Build again after changing them; CMake only recompiles what changed. Mesh resource headers (*meshres*.h*) are created by the *meshc* tool in *Orbitersdk/Utils* and go to the *build* folder.

Any IDE that opens CMake projects (for example Qt Creator, KDevelop or Visual Studio Code with the CMake extension) can open the *CMakeLists.txt* in the Orbiter *root* folder. To debug a module, run Orbiter from the *root* folder in the debugger, for example `gdb ./Orbiter`.

For what is different in Linux modules (window and drawing types, dialog resources, exported functions), see the section “Modules on Linux” in the *Orbiter Developer Manual (Linux)*.